

6	B	A and D obviously wrong. Try taking moments about C, clockwise moment = $(3m)(1.5x) = 4.5mx$ anti-clockwise moment = $(3m)(1.5x) + (m)(0.5x) = 4.5mx + 0.5mx = 5mx$ Therefore C is also wrong and centre of gravity is slightly to the left side of C.
7	C	Since the balloon is in equilibrium, upthrust on balloon = weight of balloon and helium +